MILITARY SPECIFICATION

WIRE, COPPER-CADMIUM ALLOY

This specification has been approved by the
Naval Ordnance Systems Command, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers solid, round, copper-cadmium alloy wire.

1.2 Classification

1.2.1 Temper. The alloy of wire covered by this specification shall be as furnished in the hard temper.

1.2.2 Size. The wire covered by this specification shall be nominal AWG #23, 0.0226 ±0.0002 inch in diameter.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

STANDARDS

Federal

Federal Standard No. 123	Marking for Domestic Shipment (Civilian Agencies)
Fed. Test Method Std. No. 151	Metals; Test Methods
Federal Standard No. 185	Identification Markings of Copper and Copper Base Alloy Mill Products

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 6145

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(Copies of specifications, standards, drawings, and publications required by contractors or suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 193	Resistivity Test of Electrical Conductor Material
E 8	Tension Test

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General

3.1.1 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.2 Chemical composition. The material shall conform to the chemical requirements specified in Table I, when tested in accordance with 4.4.2.

TABLE I - Chemical Requirements, Percent Composition

Copper	Cadmium	Iron
99.30 Max 98.78 Min	1.2 Max 0.7 Min	0.02 Max

3.2.1 Analysis. Analysis shall be made regularly only for the elements specifically mentioned in Table I. If, however, the presence of other elements is suspected, or indicated in the course of routine analysis, further analysis shall be made to determine the total percentage of other

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elements. A chemical composition other than that specified in Table I shall be cause for rejection of the lot from which the sample was taken.

3.3 Mechanical properties

3.3.1 Tensile properties. The tensile strength of the wire shall be 85,000 psi minimum, when tested in accordance with 4.4.3.

3.3.2 Dimensions. Wire weight will be specified in the contract, requisition or order. The wire size shall be nominal AWG #23, 0.0226 ±0.0002 inch in diameter.

3.4 Electrical properties

3.4.1 Resistivity. The resistivity shall be 12.96 ohm - circular mil/ft, maximum, when corrected to +20 degrees C (+68 degrees F) and tested in accordance with 4.4.4.

3.5 Processing requirement

3.5.1 Wire twist. No "flyer" type applicator shall be used at any stage of the wire fabrication. The wire shall be applied to or removed from a reel by rotating the reel itself.

3.6 Identification markings. When specified (see 6.2), product identification marking shall be in accordance with Federal Standard No. 185.

3.7 Workmanship. Material shall be uniform in quality and temper; clean, sound, smooth, and free from foreign material, pipes, slivers, loops, cracks, seams, scale, burrs, buckles, and the defects which, due to their nature, degree, or extent, detrimentally affect the serviceability for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified (see 6.2), a lot shall consist of 10,000 pounds, or fraction thereof, of material subjected to inspection at one time.

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4.3 Sampling. Samples taken from each lot for the purpose of the tests prescribed in this specification shall be selected in a manner that will represent correctly the material furnished to fulfill a particular contract or order.

4.3.1 For chemical analysis. The number of samples specified in Table II shall be selected from a different piece in each lot. From each sample, not less than 2 ounces of copper alloy material shall be obtained.

TABLE II - Sampling for Chemical Analysis

Pounds of Material in Lot	Number of Samples*
Up to 5000	2
5001 to 10,000	4

*If the number of original bar, billets, or cakes from which the material is processed is less than the number of samples specified, not more than one sample need be taken from each piece.

4.3.2 For mechanical examination

4.3.2.1 Tension test. Two tension test specimens shall be taken from each lot. Each specimen shall be taken from a different piece.

4.3.3 For electrical examination. Each completed length shall be used as a sample.

4.3.4 For visual and dimensional examination. From each lot of material, a representative sample shall be selected in accordance with MIL-STD-105, Inspection Level II, with an acceptable quality level (AQL) of 1.5 percent. The samples selected for dimensional examination may be the same as those selected for visual examination, but shall be evaluated separately.

4.4 Examination testing

4.4.1 Visual and dimensional. Pieces selected in accordance with 4.3.4 shall be visually examined to determine compliance with the requirements for identification marking (see 3.6) and workmanship (see 3.7), and shall be measured for compliance with the dimensional requirements (see 3.3.2) using a toolmaker's microscope or optical comparator with sufficient accuracy to ensure compliance with this specification.

4.4.2 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed by the wet chemical method in accordance with Method 111

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of Federal Test Method Standard No. 151 or the spectro-chemical method in accordance with Method 112 of Federal Test Method Standard No. 151 to determine conformance to 3.2. A single analysis of a composite sample may be made when emission methods are not used. In case of dispute, the analysis by the wet method (Method 111) shall be the basis for acceptance.

4.4.3 Tension tests. Specimens from samples selected in accordance with 4.3.2.1 shall be tested in accordance with ASTM E 8.

4.4.4 Resistivity tests. Specimens selected in accordance with 4.3.3 shall be tested in accordance with ASTM B 193.

4.5 Rejection. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.4 for that sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105.

4.5.1 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.4, subject to the provisions of the section on "Rejection and Retests" of Federal Test Method Standard No. 151.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and preservation. Not applicable.

5.2 Unit packaging and packing. The product shall be packaged and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.3 Marking (see 6.2)

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The wire is intended for use as a portion of the communication circuit between certain ordnance equipments.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Weight of reel/coil required (see 6.3).
(c) Product identification marking in accordance with Federal Standard No. 185 when required.
(d) Maximum gross weight of container.
(e) Special marking if required.
(f) Exceptions to this specification.

6.3 Density. For ordering information only, the nominal density is approximately 0.321 pound per cubic inch, which converts to approximately 1.545 pound per 1000 feet.

Custodian:
Navy - OS

Preparing Activity
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Indian Head, Maryland 20640

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-W-82598(OS) Wire, Copper-Cadmium Alloy

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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